

Model Paper - Database Management Systems

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Q1

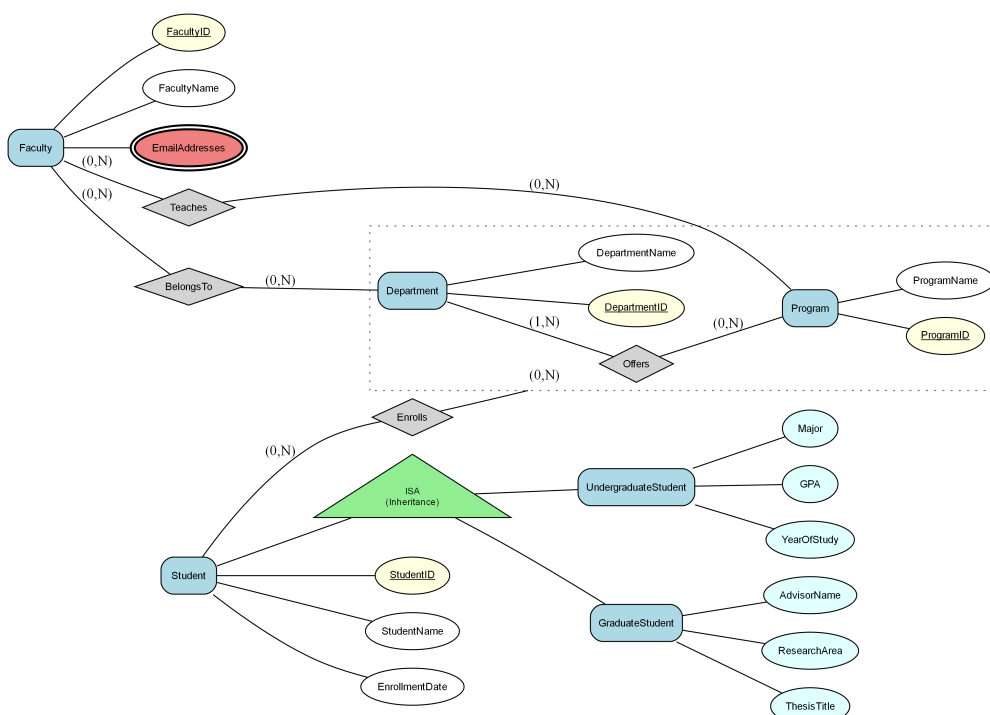
Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)
- (0,1) indicates optional participation (zero or one)
- (0,N) or (0,*) indicates optional many participation (zero or many)

In a university's course registration system, there are several entities involved: the 'Student' entity has attributes such as StudentID, Name, DateOfBirth, and EmailAddresses (where EmailAddresses is multivalued). The 'Course' entity consists of CourseID, Title, Credits, and DepartmentID. Additionally, to encapsulate the relationship between students and courses, there's an 'Enrollment' entity which tracks the EnrollmentDate and Grade received by students in specific courses. The university department, represented by the 'Department' entity, comprises DepartmentID, DepartmentName, and HeadOfDepartment, where each department offers multiple courses. A specialization exists within the 'Student' where they may be classified as either 'Graduate' or 'Undergraduate', with Graduate students having the attributes ThesisTitle and AdvisorName, while Undergraduates possess YearOfStudy and Major. Relationships should reflect the cardinality where each student can enroll in many courses, while each course can have multiple students enrolled

Figure: EER



- a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly.
- b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer.

Q2

Consider a relation $R(A, B, C, D, E)$ with the following set of functional dependencies over R : $F = \{AB \rightarrow C, B \rightarrow D, CD \rightarrow E, E \rightarrow A, A \rightarrow B\}$. This relation is used to analyze the schema to ensure it aligns with database normalization standards.

- a) Find all the keys in relation R using the attribute closure method.
- b) Is R in 3NF? Give reasons for your conclusion.
- c) Is R in BCNF? Give reasons for your conclusion. If R is not in BCNF, convert it to a set of BCNF relations.

Q3

A university is developing a comprehensive database to manage its enrollment system. The database will include several tables: "Students" with attributes (student_id: integer, name: string, email: string, major: string), "Courses" (course_id: integer, course_name: string, credits: integer), and "Enrollments" (enrollment_id: integer, student_id: integer, course_id: integer, semester: string). Each student can enroll in multiple courses, and each course can have multiple students. The university must ensure that students can only enroll in courses available during their selected semester. Additionally, students must provide a unique email address, and courses must have a defined number of credits for completion. Identify the key integrity constraints that will be necessary for this schema to maintain data integrity across these relations.

- a) In Type 2 Driver, JDBC API calls are converted to native Java API calls. Accept or refute the above statement justifying your answer.
- b)

```
String sql = "SELECT * FROM employees WHERE department = ?";
PreparedStatement pstmt = connection.prepareStatement(sql);
pstmt.setString(1, "IT");
ResultSet rs = pstmt.executeQuery();
```

There are several different statements in the JDBC API to retrieve result sets based on different requirements. Which type of statements is used in the code segment shown above? Briefly explain when this type of statement will be used.

- c) Briefly explain the options available for transferring data between different tables within the university's database schema.
- d) Briefly explain how authorization is achieved in SQL databases concerning student access to their enrollment details.
- e) **A university is establishing role-based access control for its database system that manages Courses, Students, and Enrollments. Sarah is the senior database administrator responsible for overall database administration. Emily is a junior database administrator responsible for managing the Enrollments**

database. Nathan is a database developer responsible for creating and maintaining database objects.

Michael is a data entry operator responsible for entering and updating student enrollment data.

- i. Write a T-SQL statement to create a login to Sarah with windows authentication.
- ii. Provide Sarah with the necessary permissions to handle all administrative tasks within the server using a fixed server role.
- iii. Assuming Emily's username is 'emily.e', write T-SQL statements to assign her the responsibility of managing the Enrollments database using a user-defined server role.
- iv. Assuming Nathan's login name is 'nathan.n', write T-SQL statements to grant him permission to create tables and databases.
- v. Assuming Michael's login name is 'michael.m', write T-SQL statements to allow him to perform data entry tasks on the Students and Enrollments databases.